

CMP-05 - Cognitive Orchestrator

Coordinate bounded cognitive work, model and tool routing, execution graphs, and result assembly without gaining authority to act.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 07. Authoritative state: Execution Graphs, orchestration runs, provider-selection records, result assembly state. Explicitly non-owned: Intent meaning, attention priority, judgment, capability authority, real-world commit.

Critical invariant

Orchestration coordinates work but never converts provider availability, model output, or useful results into authority.

Logical interfaces

CMP-05-IF-01 / Command / Start Orchestration

Command `Start Orchestration` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-02 / Command / Advance Execution Graph

Command `Advance Execution Graph` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-03 / Command / Cancel Run

Command `Cancel Run` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-04 / Command / Request Provider Work

Command `Request Provider Work` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-05 / Query / Get Run State

Query `Get Run State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-06 / Query / Explain Routing

Query `Explain Routing` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-07 / Query / Get Execution Graph

Query `Get Execution Graph` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-08 / Event / Run Started

Event `Run Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-09 / Event / Node Completed

Event `Node Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-10 / Event / Run Failed

Event `Run Failed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-05-IF-11 / Event / Result Assembled

Event `Result Assembled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Queued > Running > Waiting > Completed, Failed, or Cancelled. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: provider hallucination, graph deadlock, unbounded recursion, result provenance loss, cancel failure. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-05-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-05-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-05-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-05-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-05-TST-005 - Critical invariant. Orchestration coordinates work but never converts provider availability, model output, or useful results into authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-05-TST-006 - Failure containment. Inject provider hallucination, graph deadlock, unbounded recursion, result provenance loss, cancel failure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-05-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-05-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-05-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-05-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.